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Different rules for binocular combination of luminance flicker in cortical and subcortical pathways

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Abstract

1 Abstract

How does the human brain combine information across the eyes? It has been known for many years that cortical normalisation mechanisms implement ‘ocularity invariance’: equalising neural responses to spatial patterns presented either monocularly or binocularly. Here we used a novel combination of electrophysiology, psychophysics, pupillometry and computational modelling to ask whether this invariance also holds for flickering luminance stimuli with no spatial contrast. We find dramatic violations of ocularity invariance for these stimuli, both in cortex and also in the subcortical pathways that govern pupil diameter. Specifically, we find substantial binocular facilitation in both pathways with the effect being strongest in cortex. Near-linear binocular additivity (instead of ocularity invariance) was also found using a perceptual luminance matching task. Ocularity invariance is therefore not a ubiquitous feature of visual processing, and the brain appears to repurpose a generic normalisation algorithm for different visual functions by adjusting the amount of interocular suppression.

eLife assessment

This study provides potentially **important**, new information about the combination of information from the two eyes in humans. The data included frequency tagging of each eye's inputs and measures reflecting both cortical (EEG) and sub-cortical processes (pupillometry). Binocular combination is of potentially general interest because it provides -in essence- a case study of how the brain combines information from different sources and through different circuits. The strength of supporting evidence appears to be **solid**, showing that temporal modulations are combined differently than spatial modulations, with additional differences between subcortical and cortical pathways. However, the manuscript's clarity could be improved, including by adding more **convincing** motivations for the approaches used.

2 Introduction

The brain must combine information across multiple sensory inputs to derive a coherent percept of the external world. This involves a process of signal combination both within (Baker and Wade, 2017) and between (Ernst and Banks, 2002) the senses. Binocular vision is a useful test case for signal combination, as the inputs to the two eyes overlap substantially (in species with forward-facing eyes), and the neural locus is well-established (Hubel and Wiesel, 1962). Much of our knowledge about binocular combination derives from studies on the contrast response of the ‘canonical’ visual pathway, in which signals pass from the eyes to primary visual cortex (V1), via the lateral geniculate nucleus (LGN) (Purves et al., 2008). However, signals are also combined across the eyes in the network of subcortical nuclei that govern pupil diameter in response to absolute light levels (McDougal and Gamlin, 2008) and much less is known about the computations that operate in these subcortical pathways. In addition, our primary purpose here is to investigate the computations governing signal combination in these two anatomically distinct pathways in response to luminance changes.

For pattern vision, binocular presentation confers greater sensitivity to low contrast targets than monocular presentation. This is known as binocular summation, with summation ratios (the relative improvement under binocular presentation) at detection threshold lying between $\sqrt{2}$ and 2 (Baker et al., 2018; Campbell and Green, 1965). This advantage is lost at high stimulus intensities, where both psychophysical performance (contrast discrimination thresholds) (Legge, 1984; Meese et al., 2006) and neural activity (Baker and Wade, 2017; Moradi and Heeger, 2009) are approximately equal for monocular and binocular presentation. Contemporary models of binocular vision (Ding and Sperling, 2006; Meese et al., 2006) advocate a process of interocular suppression that normalizes the two eyes inputs at high contrasts and negates the binocular advantage. This is consistent with our everyday experience of ‘ocularity invariance’ (Baker et al., 2007): perceived contrast does not change when one eye is opened and closed.

The pupillary light reflex is an automatic constriction of the iris sphincter muscles in response to increases in light levels, which causes the pupil to shrink (McDougal and Gamlin, 2008). There is a clear binocular component to this reflex, as stimulation of one eye still causes constriction of the other eye’s pupil (termed the consensual response; Wyatt and Musselman (1981))). Importantly, the neuroanatomical pathway involved completely bypasses the canonical cortical pathway (retina to V1), instead involving a network of subcortical nuclei, including the Pretectal Olivary nucleus, Superior Cervical ganglion, and Edinger-Westphal nucleus (Angée et al., 2021; Mathôt, 2018; McDougal and Gamlin, 2008; Wang and Munoz, 2015). To account for the consensual response, these brain regions must combine information from the left and right eyes (Doesschate and Alpern, 1967), yet the computation that achieves this is achieved is unclear.

To address this shortcoming, we designed an experiment that allowed us to simultaneously record electrophysiological and pupillometric responses to monocular and binocular stimuli. This novel paradigm allowed us to probe both cortical (using EEG) and subcortical (using a binocular eyetracker) pathways simultaneously in response to flickering light, and make quantitative comparisons between them. Periodic flicker entrains both neural (Norcia et al., 2015) and pupil (Spitschan et al., 2014) responses at the flicker frequency, enabling precise estimation of response amplitudes in the Fourier domain. We followed up our main experiment with additional exploration of the effect of stimulus frequency, and a psychophysical matching experiment measuring perceived flicker intensity (i.e. temporal contrast). The results are interpreted using a hierarchical Bayesian computational model of binocular vision, and reveal that subcortical pathways implement stronger interocular suppression than the canonical cortical pathway.

3 Methods

3.1 Participants

Thirty (20 females), twelve (7 females) and ten (3 females) adult participants, whose ages ranged from 18 to 45, were recruited for Experiments 1, 2 and 3 respectively. All participants had normal or corrected to normal binocular vision, and gave written informed consent. Our procedures were approved by the Ethics Committee of the Department of Psychology at the University of York (identification number 792).

3.2 Apparatus & stimuli

The stimuli were two discs of achromatic flickering light with a diameter of 3.74 degrees, presented on a black background. The same stimuli were used for all three experiments. Four dark red lines were added around both discs to help with their fusion into one binocular disc (see insert in [Figure 1b](#) for an example of the fused stimulus). The discs were viewed through a four-mirror stereoscope, which used front silvered mirrors to avoid internal reflections, and meant that participants saw a single fused disc. The use of a stereoscope allowed us to modulate the stimuli in three different ocular configurations: monocular, binocular, and dichoptic. Note that during monocular presentation of flicker, the unstimulated eye still saw the static (non-flickering) disc of mean luminance.

All stimuli had a mean luminance of 42 cd/m² on an Iiyama VisionMaster™ Pro 510 display (800 × 600 pixels, 60 Hz refresh rate), which was gamma corrected using a Minolta LS-110 photometer (Minolta Camera Co. Ltd., Japan). For experiments 1 and 2, the stimuli were presented using Psychopy (v3.0.7). For experiment 3, the stimuli were presented using Psychopy (v2022.1.1).

EEG data were collected for Experiments 1 and 2 using a 64-electrode ANT WaveGuard cap and the signals were recorded at 1 kHz using the ASA software (ANT Neuro, Netherlands). Pupillometry data were collected for Experiment 1 using a binocular Pupil Core eye-tracker (Pupil Labs GmbH, Berlin, Germany; (Kassner et al. (2014))) running at 120 Hz, and the signals were recorded with the Pupil Capture software.

3.3 Procedure

Before each experiment, participants calibrated the stereoscope by adjusting the angle of the mirrors. This was done so that they would perceive the two discs as one fused disc when looking at the screen through the stereoscope.

3.3.1 Experiment 1: simultaneous EEG and pupillometry

The experiment was conducted in a windowless room, in which the only light source was the monitor. The participants sat at 99 cm from the monitor and the total optical viewing distance (through the stereoscope) was 107 cm. The experiment was carried out in a single session lasting 45 minutes in total, divided in three blocks of 15 minutes each. In each block, there were 60 trials lasting 15 seconds each (12s of stimulus presentation, with an interstimulus interval of 3s). The participants were given no task other than to look at the fixation cross in the middle of the disc while trying to minimise their blinking during the presentation period.

We included six distinct ocular conditions, each at five temporal contrast levels (combined factorially) relative to the mean luminance: 6, 12, 24, 48 and 96%. Contrast was defined as temporal Michelson contrast; the difference between maximum and minimum luminances, scaled by the mean and expressed as a percentage. In the first three conditions, the discs flickered at 2 Hz, in either a monocular, binocular, or dichoptic arrangement. In the dichoptic condition the non-target eye saw a fixed contrast of 48%. In the remaining three conditions (termed the cross-frequency conditions) one eye's disc flickered at 1.6Hz, and the other eye's disc flickered at 2Hz. We also tested monocular (one eye sees 1.6Hz flicker, the other sees mean luminance), binocular (one eye sees each frequency at the target contrast) and dichoptic (target stimulus flickering at 2Hz, mask contrast of 48% at 1.6Hz in the other eye) arrangements. The rationale for flickering both eyes at 2Hz is that we can then measure summation behaviour between the eyes in the pupil and EEG response at 2Hz. The rationale for flickering the eyes at different frequencies is that this permits measurement of suppression between the eyes (i.e. the reduction in the 2Hz response when a 1.6Hz mask component is added to the other eye). We counterbalanced presentation of the target stimulus across the left and right eyes.

3.3.2 Experiment 2: EEG responses across temporal frequency

This experiment used the same equipment set up as Experiment 1, except that the eye tracker was not used. Unlike the first experiment, only one contrast level was used (96%) and the discs were set to flicker at five different frequencies (2, 4, 8, 16 and 30 Hz). Only two ocular configurations, monocular and binocular, were included, with the latter having both discs flickering at the same frequency. The experiment was carried out in one session lasting 25 minutes in total, divided into five blocks of 5 minutes each. In each block, there were 20 trials in total with the same timing as for Experiment 1.

3.3.3 Experiment 3: temporal contrast matching

The experiment was conducted in a darkened room with a blacked-out window. The display equipment (monitor and stereoscope) were the same as for the two previous experiments, but no EEG or pupillometry data were collected. A two-interval contrast matching procedure was used to collect data. In one interval, participants were presented with a standard fused disc that flickered at a set contrast level (either 24 or 48%), which was selected by the experimenter at the beginning of each block. In the other interval, a target disc flickering at varying contrast levels was displayed. The contrast level of the target was controlled by a 1-up, 1-down staircase moving in logarithmic (dB) steps of contrast. The ratio of flicker amplitudes in the left and right eyes was varied across blocks and was set to be 0, 0.25, 0.5, 0.75 or 1 (9 distinct conditions). The standard and target discs were displayed for 1 second each, with an interstimulus interval of 0.5 seconds. After both discs had appeared on screen, the participants had to indicate which interval they perceived as having the more intense flicker. The intervals were randomly ordered, and all discs flickered at a frequency of 2 Hz (two cycles in sine phase).

Due to its long duration (approximately 3 hours in total), the participants completed the experiment across multiple sessions initiated at their own convenience. The experiment was divided into 54 blocks (3 repetitions $\times$ 2 standard contrasts $\times$ 9 target ratios), which lasted on average 3 minutes each, depending on the response speed of the participant. In each block, there were a total of 50 trials. No auditory feedback was given for this subjective task.

3.4 Data analysis

EEG data were converted from the ANT-EEProbe format to a compressed csv text file using a custom Matlab script and components of the EEGLab toolbox (Delorme and Makeig, 2004). The data for each participant were then loaded into R for analysis, where a ten-second waveform for each trial at each electrode was extracted (omitting the first two seconds). The Fourier transform of each waveform was calculated, and the complex spectrum stored in a matrix. All repetitions of each condition were then averaged for each electrode. They were then averaged across four occipital electrodes (*POz*, *Oz*, *O1*, *O2*), to obtain individual results. Finally, these were averaged across participants to obtain the group results. All averaging was performed in the complex domain and therefore retained the phase information (i.e. coherent averaging), and at each stage we excluded data points with a Mahalanobis distance exceeding $D = 3$ from the complex-valued mean (see (Baker, 2021)). For statistical comparisons of complex-valued data, we use the $ANOV A^2_{circ}$ statistic described by (Baker (2021)). This is a multivariate extension of ANOVA that assumes equal variance of the real and imaginary Fourier components, or equivalently, an extension of the T^2_{circ} statistic of (Victor and Mast (1991)) that can compare more than two conditions.

A similar analysis pipeline was adopted for the pupillometry data. The data were converted from mp4 videos to a csv text file using the Pupil Player software (Kassner et al., 2014), which estimated pupil diameter for each eye on each frame using a 3D model of the eyeball. The individual data were then loaded into R for analysis, where again a ten-second waveform for each trial in each eye was extracted (excluding the first two seconds after stimulus onset). We interpolated across any dropped or missing frames to ensure regular and continuous sampling over time. The Fourier transform was calculated for each waveform, and all repetitions of each condition were pooled across eye and then averaged. Finally, data were averaged across all participants to obtain the group results. Again, we used coherent averaging, and excluded outlying data points in the same way as for the EEG data. Note that previous pupillometry studies using luminance flicker have tended to fit a single sine-wave at the fundamental frequency, rather than using Fourier analysis (e.g. (Spitschan et al., 2014)). The Fourier approach is more robust to noise at other frequencies and has been used in some previous studies (see (Barrionuevo et al., 2014); (Barrionuevo and Cao, 2016)). Additionally, it makes the pupillometry analysis consistent with standard practice in steady-state EEG analysis (e.g. (Figueira et al., 2022)).

To analyse the matching data, we pooled the trial responses across all repetitions of a given condition for each participant. We then fitted a cumulative normal psychometric function to estimate the point of subjective equality at the 50% level. Thresholds were averaged across participants in logarithmic (dB) units.

For all experiments, we used a bootstrapping procedure with 1000 iterations to estimate standard errors across participants. All analysis and figure construction was conducted using a single R-script, available online, making this study fully computationally reproducible.

3.5 Computational model and parameter estimation

To describe our data, we chose a model of binocular contrast gain control with the same general form as the first stage of the model proposed by (Meese et al. (2006)). The second gain control stage was omitted (consistent with (Baker and Wade, 2017)) to simplify the model and reduce the number of free parameters. The response of the left eye's channel is given by:

$$Resp_L = \frac{L^2}{Z + L + wR}, \quad (1)$$

with an equivalent expression for the right eye:

$$Resp_R = \frac{R^2}{Z + R + wL}. \quad (2)$$

In both equations, L and R are the contrast signals from the left and right eyes, Z is a saturation constant that shifts the contrast-response function laterally, and w is the weight of suppression from the other eye.

The responses from the two eyes are then summed binocularly:

$$Resp_B = R_{max}(Resp_L + Resp_R) + n, \quad (3)$$

where n is a noise parameter, and R_{max} scales the overall response amplitude. The R_{max} parameter was omitted when modelling the contrast matching data, as it has no effect in this paradigm.

Despite being derived from the model proposed by (Meese et al. (2006)), the simplifications applied to this architecture make it very similar to other models (e.g. (Ding and Sperling, 2006); (Doesschate and Alpern, 1967); (Legge, 1984); (Schrödinger, 1926)). In particular we fixed the numerator exponent at 2 in our model, because otherwise this value tends to trade off with the weight of interocular suppression (see (Baker et al., 2012); (Kingdom and Libenson, 2015)). Our key parameter of interest is the weight of interocular suppression. Large values around $w = 1$ result in a very small or nonexistent binocular advantage at suprathreshold contrasts, consistent with previous work using grating stimuli (Baker and Wade, 2017). Low values around $w = 0$ produce substantial, near-linear binocular facilitation (Baker et al., 2020).

We implemented the model within a Bayesian framework using the Stan software (Carpenter et al., 2017). This allowed us to estimate group-level posterior parameter distributions for the weight of interocular suppression, w , and the other free model parameters R_{max} , Z and n . The prior distributions for all parameters were Gaussian, with means and standard deviations of 1 and 0.5 for w and R_{max} , and 5 and 2 for Z and n , with these values chosen based on previous literature (Baker et al., 2012; Meese et al., 2006). We sampled from a Student's t-distribution for the amplitudes in the pupillometry and EEG experiments, and from a Bernoulli distribution for the single trial matching data. The models were fit using the individual data across all participants, independently for each data set. Because we used coherent averaging across participants, the group average amplitudes are shifted vertically relative to the model predictions, which are based on hierarchical fits to the individual participant amplitudes (put another way, the model does not implement coherent averaging). However it is clear that the character of the model in all cases gives a good representation of the data, as can be seen in Figure 6. We took posterior samples at over a million steps for each data set, using a computer cluster, and retained 10% of samples for plotting.

3.6 Preregistration, data and code availability

We initially preregistered our main hypotheses and analysis intentions for the first experiment. We then conducted a pilot study with $N=12$ participants, before making some minor changes to the stimulus (we added dim red lines to aid binocular fusion). We then ran the main experiment, followed by two additional experiments that were not preregistered. The preregistration document, raw data files, and experimental and analysis code, are available on the project repository: 2023 eLife. <https://doi.org/10.17605/OSF.IO/TBEMA>.

4 Results

4.1 Experiment 1

The pupillometry results are summarised in Figure 1. The group average waveform for binocular presentation is shown in Figure 1a. There is a substantial pupil constriction at stimulus onset, followed by visible oscillations at the flicker frequency (2Hz, see waveform at foot). The average Fourier spectrum is displayed in Figure 1b, and shows a clear spike at 2 Hz, but no evidence of a second harmonic response at 4Hz. These results demonstrate that our paradigm can evoke measurable steady-state pupil responses at 2Hz.

Pupillometry

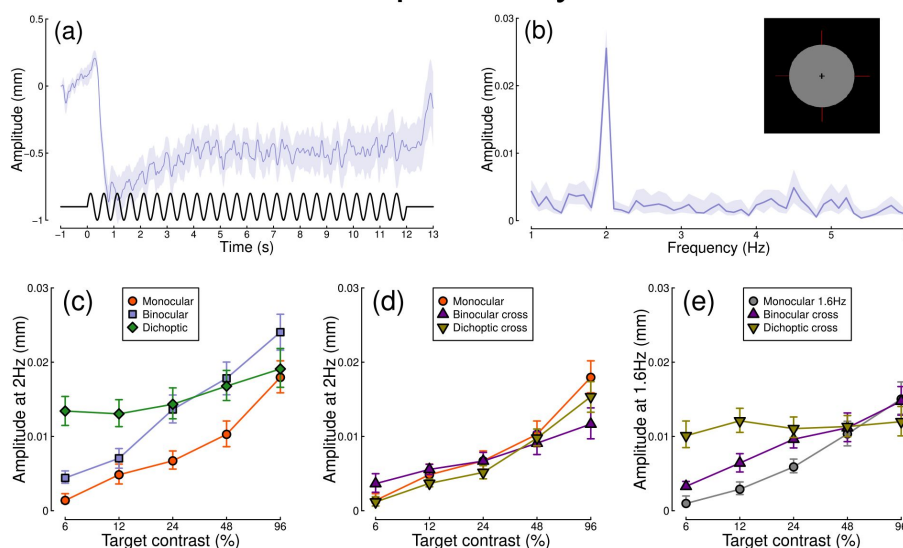


Figure 1:

Summary of pupillometry results for $N=30$ participants. Panel (a) shows a group average waveform for binocular presentation (low pass filtered at 5Hz), with the driving signal plotted at the foot. Panel (b) shows the average Fourier spectrum, with an inset image illustrating the stimulus appearance (upper right). Panels (c,d) show contrast response functions at 2Hz for different conditions. Panel (e) shows contrast response functions at 1.6Hz for three conditions. Shaded regions and error bars indicate bootstrapped standard errors.

Figure 1c shows contrast response functions in response to stimuli flickering only at 2Hz. Response amplitudes increased monotonically with target contrast, confirming that our paradigm is suitable for measuring contrast-dependent differences in response (to our knowledge this is the first time this has been demonstrated). The amplitude of the binocular condition (blue squares) is consistently greater than that of the monocular condition (red circles) across all target contrasts. A 2×5 repeated measures ANOVA²_{circ} (Baker, 2021) comparing these conditions revealed a significant main effect of target contrast ($F(8,580) = 16.79$, $p < 0.001$), a significant effect of condition ($F(2,580) = 11.04$, $p < 0.001$), and a significant interaction ($F(8,580) = 56.25$, $p < 0.001$). The dichoptic condition begins at a much higher amplitude, owing to binocular combination of the target and high (48%) contrast mask, and

then increases slightly with increasing target contrast (main effect of target contrast: $F(8,232) = 3.03$, $p < 0.003$).

In Figure 1d, we plot responses to monocular target stimuli flickering at 2Hz, when the other eye viewed stimuli flickering at 1.6Hz (the red monocular-only data are replotted from Figure 1c for comparison). When the 1.6Hz component had the same contrast as the target (the binocular cross condition, shown in purple) responses were facilitated slightly at low contrasts, and suppressed at the highest target contrasts (interaction between contrast and condition: $F(8,580) = 52.94$, $p < 0.001$). When the 1.6Hz component had a fixed contrast of 48% (the dichoptic cross condition, shown in yellow), responses were suppressed slightly across the contrast range (interaction between contrast and condition: $F(8,580) = 62.05$, $p < 0.001$).

Figure 1e shows responses at 1.6Hz, for the same conditions, as well as for a condition in which a monocular stimulus flickered at 1.6Hz (grey circles). Surprisingly there again appears to be a slight facilitation effect in the binocular cross condition, particularly at lower contrasts. The dichoptic cross condition does not show clear modulation with target contrast.

Figure 2 shows equivalent results, measured contemporaneously using EEG. Figure 2a shows the group average waveform for binocular presentation, and Figure 2b shows the Fourier spectrum for binocular presentation, both averaged across four posterior electrodes (Oz, POz, O1 and O2, marked on the inset scalp plots). Unlike for the pupillometry data, there are clear responses at both the first harmonic frequency (2Hz), and also the second harmonic frequency (4Hz). We therefore calculated contrast response functions at both first and second harmonic frequencies.

Electroencephalography

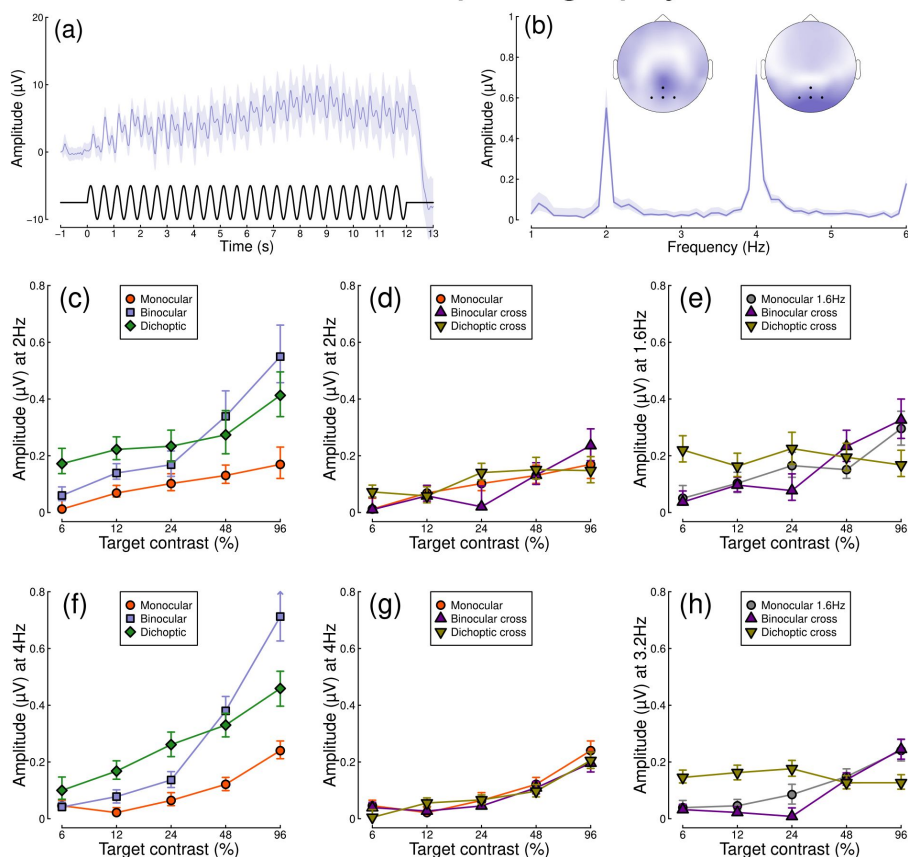


Figure 2:

Summary of EEG results for N=30 participants. Panel (a) shows a group average waveform for binocular presentation (low pass filtered at 5Hz), with the driving signal plotted at the foot. Panel (b) shows the average Fourier spectrum, and inset scalp distributions. Black dots on the scalp plots indicate electrodes Oz, POz, O1 and O2. Panels (c,d) show contrast response functions at 2Hz for different conditions. Panel (e) shows contrast response functions at 1.6Hz for three conditions. Panels (f-h) are in the same format but for the second harmonic responses. Shaded regions and error bars indicate bootstrapped standard errors.

When stimuli in both eyes flicker at 2Hz, the binocular responses at the first (Figure 2c) and second (Figure 2f) harmonics are substantially greater than the monocular responses, particularly at high contrasts. Analysis of variance on the complex values ($ANOV A_{circ}^2$) revealed a main effect of contrast ($F(8,580) = 4.38, p < 0.001$) and an interaction effect ($F(8,580) = 61.58, p < 0.001$), but no effect of condition ($p = 0.13$) at the first harmonic, with a similar pattern of results obtained at the second harmonic. For the cross-frequency conditions (Figure 2d,g), there was no appreciable effect of adding a 1.6Hz component on the response at 2Hz or 4Hz (no effect of condition, and no interaction). Similarly, there were no clear interocular interactions between frequencies in the responses at 1.6Hz (Figure 2e) and 3.2Hz (Figure 2h). This pattern of results suggests that processing of temporal luminance modulations happens in a more linear way in visual cortex (indexed by EEG), compared with subcortical pathways (indexed by pupillometry), and shows no evidence of interocular suppression.

Finally, we calculated the ratio of binocular to monocular responses across the three data types from Experiment 1. Figure 3 shows that these ratios are approximately $\sqrt{2}$ across the low-to-intermediate contrast range for all three data types. At higher contrasts, we see ratios of 2 or higher for the EEG data, but much weaker ratios near 1 for the pupillometry data. Note that the ratios here are calculated on a per-participant basis and then averaged, rather than being the ratios of the average values shown in Figures 1 and 2. A 3×5 repeated measures ANOVA on the logarithmic (dB) ratios found a main effect of contrast ($F(3.08,89.28) = 4.53, p < 0.002$), no effect of data modality ($F(2,58) = 0.75, p = 0.48$), but a highly significant interaction ($F(5.54,160.67) = 3.84, p < 3 \times 10^{-4}$).

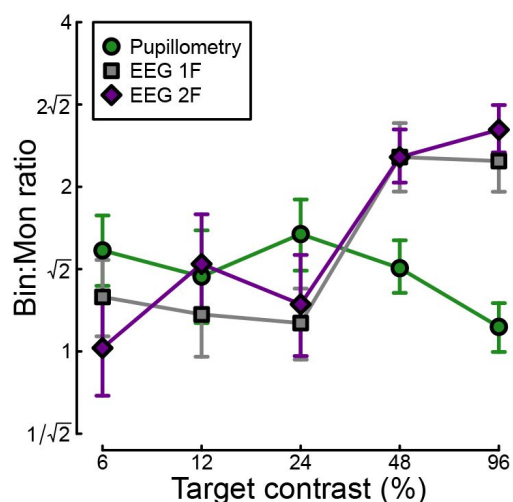


Figure 3:

Ratio of binocular to monocular response for three data types. Each ratio is the average of ratios for N=30 participants, and error bars indicate bootstrapped standard errors.

4.2 Experiment 2

The strong binocular facilitation and weak interocular suppression in the EEG data from Experiment 1 was very different from previous findings on binocular combination using steady-state EEG with grating stimuli (Baker and Wade, 2017). One possible explanation is that the lower temporal frequency used here (2Hz, vs 5 or 7Hz in previous work) might be responsible for this difference. We therefore ran a second experiment to compare monocular and binocular responses at a range of temporal frequencies. Only EEG data were collected for this experiment, as the pupil response is negligible above around 2Hz (Spitschan et al., 2014); note that we originally chose 2Hz because it produces measurable signals for both EEG and pupillometry, yet is unfortunately optimal for neither.

Results from the temporal frequency experiment are shown in Figure 4. Figure 4a shows the Fourier spectra for responses to binocular flicker at 5 different frequencies (2, 4, 8, 16, and 30 Hz). From 2 to 16 Hz, clear signals are observed at each fundamental frequency, and typically also their higher harmonics (integer multiples of the fundamental). However, at 30 Hz (upper row), the responses recorded were not demonstrably above the noise baseline. Figure 4b compares the monocular and binocular responses at each stimulation frequency. Here we replicate the substantial summation effect across frequencies up to and including 16 Hz (Fig. 4c), demonstrating that strong binocular facilitation in the EEG data of Experiment 1 cannot be attributed to our use of 2 Hz flicker.

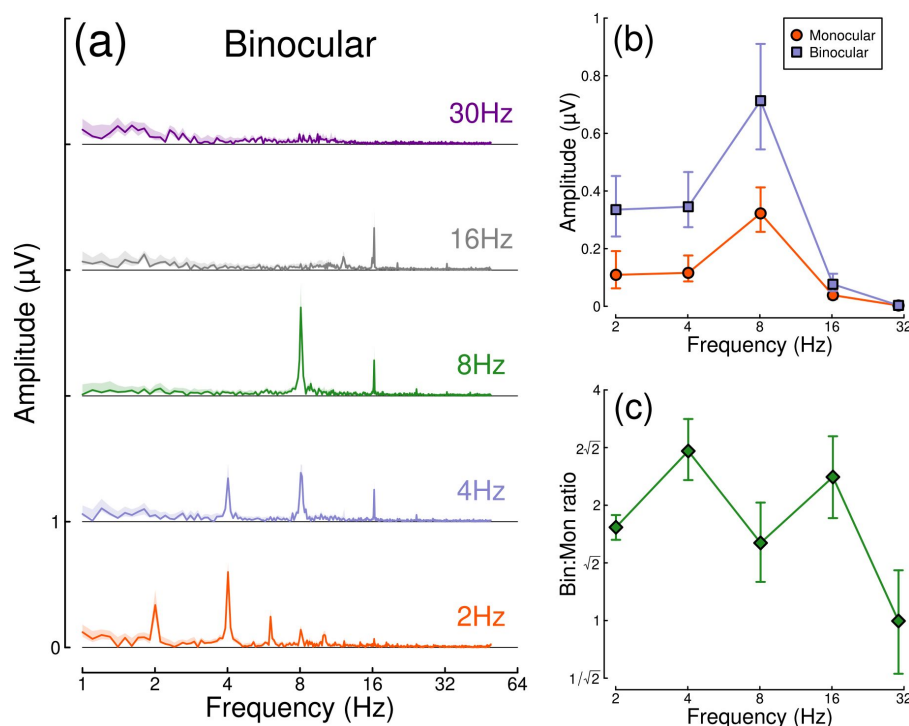


Figure 4:

Binocular facilitation at different temporal frequencies. Panel (a) shows Fourier spectra for responses to binocular flicker at 5 different frequencies (offset vertically for clarity). Panel (b) shows the response at each stimulation frequency for monocular (red circles) and binocular (blue squares) presentation. Panel (c) shows the ratio of binocular to monocular responses. Error bars and shaded regions indicate bootstrapped standard errors across $N=12$ participants.

4.3 Experiment 3

In Experiment 1 we found evidence of stronger binocular facilitation for cortical responses to luminance flicker (measured using EEG), compared with subcortical responses (measured using pupillometry). Since perception is dependent on cortical responses, these results provide a clear prediction for perceived contrast judgements indexed by psychophysical contrast matching paradigms (e.g. (Anstis and Ho, 1998); (Levelt, 1965); (Quaia et al., 2018)). We therefore conducted such an experiment, in which participants judged which of two stimuli had the greater perceived amplitude of flicker. One stimulus was a matching stimulus, that had a fixed binocular flicker amplitude of either 24% or 48% (temporal) contrast. The other stimulus was a target stimulus, the contrast of which was controlled by a staircase algorithm. We tested 9 ratios of contrast between the left and right eyes.

The results from the matching experiment are shown in Figure 5. Each data point indicates the contrast levels required in each eye that were perceptually equivalent to the binocular 24% (red circles) and 48% (blue circles) matching contrasts. At both matching contrasts, we see a very substantial increase in the physical contrast required for a monocular target (data points along the x- and y-axes), compared to a binocular target (points along the diagonal of $x=y$). For example with a 48% match, the monocular targets required contrasts close to 100%, whereas binocular targets required a contrast of around 50%. The data points between these

extremes also fall close to the predictions of a linear summation model (diagonal dotted lines), and are inconsistent with a winner-takes-all (or MAX) model (dashed lines). Overall, these matching results are consistent with the approximately linear summation effects observed in the EEG data of Experiment 1 (Figure 2c,f).

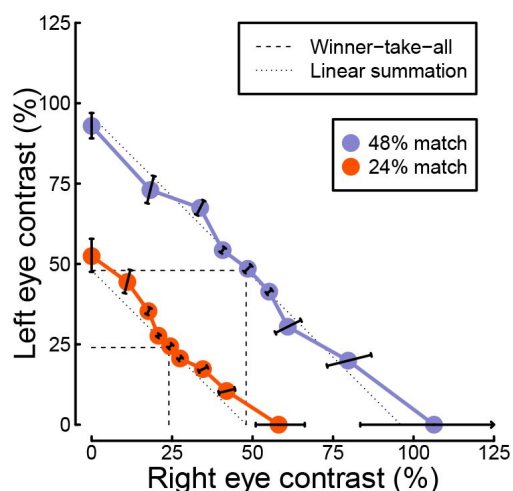


Figure 5:

Contrast matching functions. Dotted and dashed lines are predictions of canonical summation models with a linear exponent (dotted) or an infinite exponent (dashed). Error bars indicate the standard error across participants ($N=10$), and are constrained along radial lines converging at the origin. Note that, for the 48% match, the data point on the x axis falls higher than 100% contrast. This is because the psychometric function fits for some individuals were interpolated such that the PSE fell above 100%, shifting the mean slightly above that value.

4.4 Computational modelling

We fitted a computational model to the data from Experiments 1 & 3 using a hierarchical Bayesian approach. The model behaviour is displayed in Figure 6e-h, with empirical data replotted in Figure 6a-d for comparison. In general, the model captures the key characteristics of the empirical data. Note that there are some minor discrepancies, which are a consequence of the hierarchical nature of the modelling. In brief, the model is fitted to the amplitudes for each participant, and group-level parameter estimates are derived based on these fits (see Table 1). This procedure discards the phase information, whereas the empirical averages are coherently averaged across participants (retaining phase information). This explains the amplitude differences between model and data, particularly at low target contrast levels, but is of little consequence for the pattern of relative responses across conditions, which is our main focus here.

Table 1:

Summary of median parameter values.

Data set	Z	k	w	Rmax
Pupillometry	2.37	0.01	0.67	0.00023
EEG 1F	2.59	0.15	0.02	0.0026
EEG 2F	2.21	0.06	0.01	0.00404
Matching	0.30	5.07	0.09	-

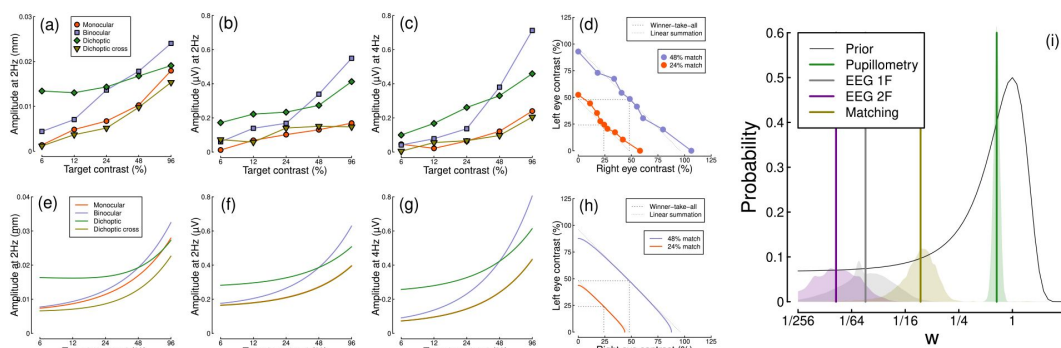


Figure 6:

Summary of computational modelling. Panels (a-d) show empirical data from key conditions, replotted from earlier figures for the pupillometry (a), first harmonic EEG responses (b), second harmonic EEG responses (c) and contrast matching (d) experiments. Panels (e-h) show model behaviour for the same conditions, generated using the median group-level parameter values. Panel (i) shows the posterior probability distributions of the interocular suppression parameter for each of the four model fits. The pupillometry distribution (green) is centred about a substantially higher suppressive weight than for the other data types (note the logarithmic x-axis). The black curve shows the (scaled) prior distribution for the weight parameter.

We were particularly interested in comparing the weight of interocular suppression across data sets. We therefore plot the posterior distributions for this parameter for all four data sets (see Figure 6i). The key finding is that the pupillometry results (green distribution) display a much greater weight of interocular suppression compared with the other data sets (grey, purple and yellow distributions). There is no overlap between the pupillometry distribution and any of the other three. All four distributions are also meaningfully below a weight of 1 – the value that previous work using grating stimuli would predict (Baker and Wade, 2017; Meese et al., 2006), and the peak location of our prior distribution (black curve). These results offer an explanation of the empirical data: the strong interocular suppression for the pupillometry data is consistent with the weak binocular facilitation, and measurable dichoptic masking observed using that method. The weaker suppression for the other experiments is consistent with the near-linear binocular facilitation effects, and absent dichoptic masking.

5 Discussion

Using a novel paradigm that combines EEG and pupillometry, we found surprising results for the binocular integration of flickering light. In the visual cortex, the processing of spatially-uniform temporal luminance modulations seems to happen approximately linearly, with no evidence of interocular suppression. In the subcortical pathway, signal combination is more non-linear, with evidence of interocular suppression. This pattern of results was confirmed by computational modelling, which showed a much greater suppressive weight for the pupillometry data compared to the EEG data. Additionally, we found that perception of flickering light is consistent with a near-linear summation process, consistent with the cortical (EEG) responses.

The results of our main experiment were unexpected for both the pupillometry and the EEG measures. Previous studies investigating binocular combination of spatial patterns (i.e. sine wave grating stimuli) are all consistent with strong interocular suppression and weak binocular facilitation at high contrasts (Baker and Wade, 2017; Meese et al., 2006; Moradi and Heeger, 2009). Our second experiment ruled out the possibility that these differences were due to the lower temporal frequency (2Hz) used here. However, there is evidence of more extensive binocular facilitation for a range of other stimuli. For example, (Quaia et al. (2018)) observed a strong binocular facilitation (or ‘supersummation’) in the reflexive eye movement response to rapidly moving stimuli, and (Spitschan and Cajochen (2019)) report a similar result in archival data on melatonin suppression due to light exposure. In the auditory system, interaural suppression of amplitude modulation also appears to be weak when measured using a similar steady-state paradigm (Baker et al., 2020). Finally, psychophysical matching experiments using static stimuli also show near-linear behaviour for luminance increments (Anstis and Ho, 1998; Baker et al., 2012; Levelt, 1965), though not for luminance decrements (Anstis and Ho, 1998). Overall, this suggests that strong interocular normalization may be specific to spatial pattern vision, and not a general feature of sensory signal combination.

Given the above, where does this leave our understanding of the overarching purpose of signal combination? (Baker and Wade (2017)) point out that strong suppression between channels that are subsequently summed is equivalent to a Kalman filter, which is the optimal method for combining two noisy inputs (see also (Ernst and Banks, 2002)). This account has intuitive appeal, and is consistent with other models that propose binocular combination as a means of redundancy reduction (Li and Atick, 1994; May and Zhaoping, 2022). One possibility is that optimal combination is useful for visual perception — a critical system for interacting with the local environment — and is therefore worth devoting the additional resource of inhibitory wiring between ocular channels. However the other examples given in the previous paragraph are primarily physiological responses (pupil size, eye movements, hormone release) that may benefit less from an increased signal-to-noise ratio, or otherwise be phylogenetically older than binocular pattern vision. Conceptualised another way, the brain can repurpose a generic architecture for different situational demands by adjusting parameter values (here the weight of interocular suppression) to achieve different outcomes. Our future work in this area intends to compare binocular combination for specific photoreceptor pathways, including different cone classes, and intrinsically photoreceptive retinal ganglion cells.

Pupil size determines the total amount of light falling on the retina. It is therefore the case that fluctuations in pupil diameter have a downstream effect on the signals reaching cortex. We did not incorporate such interactions into our computational model, though in principle this might be worthwhile. However we anticipate that any such effects would be small, since pupil modulations at 2Hz are in the order of 2% of overall diameter (e.g. (Spitschan et al., 2014)). It is also the case that cortical activity can modulate pupil diameter, usually through arousal and attention mechanisms (e.g. (Bradley et al., 2008)). We think it unlikely that these temporally coarse processes would have a differential effect on e.g. monocular and binocular stimulation conditions in our experiment, and any fluctuations during an experimental session (perhaps owing to fatigue) will be equivalent for our comparisons of interest. Therefore we make the simplifying assumption that the two pathways are effectively distinct, but hope to investigate this directly in future neuroimaging work.

Classic studies investigating the neurophysiological architecture of V1 reported that cells in cytochrome-oxidase ‘blobs’ (Horton and Hubel, 1981; Livingstone and Hubel, 1984) are biased towards low spatial frequencies (Edwards et al., 1995; Tootell et al., 1988), and relatively insensitive to stimulus orientation (Horton and Hubel, 1981; Livingstone and Hubel, 1984; Economides et al., 2011). As the blob regions are embedded within ocular dominance

columns (Horton and Hubel, 1981), they are also largely monocular (Livingstone and Hubel, 1984; Tychsen et al., 2004). More recent work has reported psychophysical evidence for unoriented chromatic (Gheiratmand et al., 2013) and achromatic (Meese and Baker, 2011) mechanisms, that also appear to be monocular. Our use of luminance flicker might preferentially stimulate these mechanisms, perhaps explaining why our EEG data show little evidence of binocular interactions. Indeed, our EEG results could potentially be explained by a model involving entirely non-interacting monocular channels, with the binocular facilitation effects we find (e.g. [Figures 3 & 4](#)) owing to additivity of the electrophysiological response across independent monocular cells, rather than requiring binocular neurons. However our matching data ([Figure 5](#)), as well as everyday perceptual experience, indicates that luminance signals must be combined physiologically at some stage, and that this process still involves weak or absent interocular suppression.

6 Conclusions

We have demonstrated that binocular combination of flickering light differs between cortical and subcortical pathways. Flicker was also associated with substantially weaker interocular suppression, and stronger binocular facilitation, compared to combination of spatial luminance modulations in visual cortex. Our computational framework for understanding signal combination permits direct comparisons between disparate experimental paradigms and data types. We anticipate that this will help elucidate the constraints the brain faces when combining different types of signals to govern perception, action and biological function.

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8 Author contributions

Federico Segala: Methodology, software, formal analysis, investigation, data curation, writing - original draft, writing - review & editing, visualization. **Aurelio Bruno:** Conceptualization, writing - review & editing, supervision, project administration, funding acquisition. **Myat Aung:** Software, resources, writing - review & editing. **Alex Wade:** Conceptualization, methodology, resources, writing - review & editing, supervision, project administration, funding acquisition. **Daniel Baker:** Conceptualization, methodology, software, formal analysis, investigation, resources, data curation, writing - original draft, writing - review & editing, visualization, supervision, project administration, funding acquisition.

9 Declaration of interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

In this paper, the interocular/binocular combination of temporal luminance modulations is studied. Binocular combination is of broad interest because it provides a remarkable case study of how the brain combines information from different sources. In addition, the mechanisms of binocular combination are of interest to vision scientists because they provide insight into when/where/how information from two eyes is combined.

This study focuses on how luminance flicker is combined across two eyes, extending previous work that focused mainly on spatial modulations. The results appear to show that temporal modulations are combined in different ways, with additional differences between subcortical and cortical pathways.

1. Main concern: subcortical and cortical pathways are assessed in quite different ways. On the one hand, this is a strength of the study (as it relies on unique ways of interrogating each

pathway). However, this is also a problem when the results from two approaches are combined - leading to a sort of attribution problem: Are the differences due to actual differences between the cortical and subcortical binocular combinations, or are they perhaps differences due to different methods. For example, the results suggest that the subcortical binocular combination is nonlinear, but it is not clear where this nonlinearity occurs. If this occurs in the final phase that controls pupillary responses, it has quite different implications.

At the very least, this work should clearly discuss the limitations of using different methods to assess subcortical and cortical pathways.

2. Adding to the previous point, the paper needs to be a better job of justifying not only the specific methods but also other details of the study (e.g., why certain parameters were chosen). To illustrate, a semi-positive example: Only page 7 explains why 2Hz modulation was used, while the methods for 2Hz modulation are described in detail on page 3. No justifications are provided for most of the other experimental choices. The paper should be expanded to better explain this area of research to non-experts. A notable strength of this paper is that it should be of interest to those not working in this particular field, but this goal is not achieved if the paper is written for a specialist audience. In particular, the introduction should be expanded to better explain this area of research, the methods should include justifications for important empirical decisions, and the discussion should make the work more accessible again (in addition to addressing the issues raised in point 1 above). The results also need more context. For example, why EEG data have overtones but pupillometry does not?

Reviewer #2 (Public Review):

Previous studies have extensively explored the rules by which patterned inputs from the two eyes are combined in the visual cortex. Here the authors explore these rules for unpatterned inputs (luminance flicker) at both the level of the cortex, using Steady-State Visual Evoked Potentials (SSVEPs) and at the sub-cortical level using pupillary responses. They find that the pattern of binocular combination differs between cortical and sub-cortical levels with the cortex showing less dichoptic masking and somewhat more binocular facilitation.

Importantly, the present results with flicker differ markedly from those with gratings (Hou et al., 2020, J Neurosci, Baker and Wade 2017 cerebral cortex, Norcia et al, 2000 Neuroreport, Brown et al., 1999, IOVS). When SSVEP responses are measured under dichoptic conditions where each eye is driven with a unique temporal frequency, in the case of grating stimuli, the magnitude of the response in the fixed contrast eye decreases as a function of contrast in the variable contrast eye. Here the response increases by varying (small) magnitudes. The authors favor a view that cortex and perception pool binocular flicker inputs approximately linearly using cells that are largely monocular. The lack of a decrease below the monocular level when modulation strength increase is taken to indicate that previously observed normalization mechanism in pattern vision does not play a substantial role in the processing of flicker. The authors present a computational model of binocular combination that captures features of the data when fit separately to each data set. Because the model has no frequency dependence and is based on scalar quantities, it cannot make joint predictions for the multiple experimental conditions which is one of its limitations.

A strength of the current work is the use of frequency-tagging of both pupil and EEG responses to measure responses for flicker stimuli at two anatomical levels of processing. Flicker responses are interesting but have been relatively neglected. The tagging approach allows one to access responses driven by each eye, even when the other eye is stimulated which is a great strength. The tagging approach can be applied at both levels of processing at

the same time when stimulus frequencies are low, which is an advantage as they can be directly compared. The authors demonstrate the versatility of frequency tagging in a novel experimental design which may inspire other uses, both within the present context and others. A disadvantage of the tagging approach for studying sub-cortical dynamics via pupil responses is that it is restricted to low temporal frequencies given the temporal bandwidth of the pupil. The inclusion of a behavioral measure and a model is also a strength, but there are some limitations in the modeling (see below).

The authors suggest in the discussion that luminance flicker may preferentially drive cortical mechanisms that are largely monocular and in the results that they are approximately linear in the dichoptic cross condition (no effect of the fixed contrast stimulus in the other eye). By contrast, prior research using dichoptic dual frequency flickering stimuli has found robust intermodulation (IM) components in the VEP response spectrum (Baitch and Levi, 1988, *Vision Res*; Stevens et al., 1994 *J Ped Ophthal Strab*; France and Ver Hoeve, 1994, *J Ped Ophthal Strab*; Suter et al., 1996 *Vis Neurosci*). The presence of IM is a direct signature of binocular interaction and suggests that at least under some measurement conditions, binocular luminance combination is "essentially" non-linear, where essential implies a point-like non-linearity such as squaring of excitatory inputs. The two views are in striking contrast. It would thus be useful for the authors could show spectra for the dichoptic, two-frequency conditions to see if non-linear binocular IM components are present.

If the IM components are indeed absent, then there is a question of the generality of the conclusions, given that several previous studies have found them with dichoptic flicker. The previous studies differ from the authors' in terms of larger stimuli and in their use of higher temporal frequencies (e.g. 18/20 Hz, 17/21 Hz, 6/8 Hz). Either retinal area stimulated (periphery vs central field) or stimulus frequency (high vs low) could affect the results and thus the conclusions about the nature of dichoptic flicker processing in cortex. It would be interesting to sort this out as it may point the research in new directions.

Whether these components are present or absent is of interest in terms of the authors' computational model of binocular combination. It appears that the present model is based on scalar magnitudes, rather than vectors as in Baker and Wade (2017), so it would be silent on this point. The final summation of the separate eye inputs is linear in the model. In the first stage of the model, each eye's input is divided by a weighted input from the other eye. If we take this input as inhibitory, then IM would not emerge from this stage either.

Related to the model: One of the more striking results is the substantial difference between the dichoptic and dichoptic-cross conditions. They differ in that the latter has two different frequencies in the two eyes while the former has the same frequency in each eye. As it stands, if fit jointly on the two conditions, the model would make the same prediction for the dichoptic and dichoptic-cross conditions. It would also make the same prediction whether the two eyes were in-phase temporally or in anti-phase temporally. There is no frequency/phase-dependence in the model to explain differences in these cases or to potentially explain different patterns at the different VEP response harmonics. The model also fits independently to each data set which weakens its generality. An interpretation outside of the model framework would thus be helpful for the specific case of differences between the dichoptic and dichoptic-cross conditions.

Prior work has defined several regimes of binocular summation in the VEP (Apkarian et al., 1981 *EEG Journal*). It would be useful for the authors to relate the use of their terms "facilitation" and "suppression" to these regimes and to justify/clarify differences in usage, when present. Experiment 1, Fig. 3 shows cases where the binocular response is more than twice the monocular response. Here the interpretation is clear: the responses are super-additive and would be classed as involving facilitation in the Apkarian et al framework.

In the Apkarian et al framework, a ratio of 2 indicates independence/linearity. Ratios between 1 and 2 indicate sub-additivity and are diagnostic of the presence of binocular interaction but are noted by them to be difficult to interpret mechanistically. This should be discussed. A ratio of <1 indicates frank suppression which is not observed here with flicker.

Can the model explore the full range of binocular/monocular ratios in the Apkarian et al framework? I believe much of the data lies in the "partial summation" regime of Apkarian et al and that the model is mainly exploring this regime and is a way of quantifying varying degrees of partial summation.

Reviewer #3 (Public Review):

This manuscript describes interesting experiments on how information from the two eyes is combined in cortical areas, sub-cortical areas, and perception. The experimental techniques are strong and the results are potentially quite interesting. But the manuscript is poorly written and tries to do too much in too little space. I had a lot of difficulty understanding the various experimental conditions, the complicated results, and the interpretations of those results. I think this is an interesting and useful project so I hope the authors will put in the time to revise the manuscript so that regular readers like myself can better understand what it all means.

Now for my concerns and suggestions:

The experimental conditions are novel and complicated, so readers will not readily grasp what the various conditions are and why they were chosen. For example, in one condition different flicker frequencies were presented to the two eyes (2Hz to one and 1.6Hz to the other) with the flicker amplitude fixed in the eye presented to the lower frequency and the flicker amplitude varied in the eye presented to the higher frequency. This is just one of several conditions that the reader has to understand in order to follow the experimental design. I have a few suggestions to make it easier to follow. First, create a figure showing graphically the various conditions. Second, come up with better names for the various conditions and use those names in clear labels in the data figures and in the appropriate captions. Third, combine the specific methods and results sections for each experiment so that one will have just gone through the relevant methods before moving forward into the results. The authors can keep a general methods section separate, but only for the methods that are general to the whole set of experiments.

I wondered why the authors chose the temporal frequencies they did. Barrionuevo et al (2014) showed that the human pupil response is greatest at 1Hz and is nearly a log unit lower at 2Hz (i.e., the change in diameter is nearly a log unit lower; the change in area is nearly 2 log units lower). So why did the authors choose 2Hz for their primary frequency? And why did the authors choose 1.6Hz which is quite close to 2Hz for their off frequency? The rationale behind these important decisions should be made explicit.

By the way, I wondered if we know what happens when you present the same flicker frequencies to the two eyes but in counter-phase. The average luminance seen binocularly would always be the same, so if the pupil system is linear, there should be no pupil response to this stimulus. An experiment like this has been done by Flitcroft et al (1992) on accommodation where the two eyes are presented stimuli moving oppositely in optical distance and indeed there was no accommodative response, which strongly suggests linearity.

Figures 1 and 2 are important figures because they show the pupil and EEG results, respectively. But it's really hard to get your head around what's being shown in the lower

row of each figure. The labeling for the conditions is one problem. You have to remember how "binocular" in panel c differs from "binocular cross" in panel d. And how "monocular" in panel d is different than "monocular 1.6Hz" in panel e. Additionally, the colors of the data symbols are not very distinct so it makes it hard to determine which one is which condition. These results are interesting. But they are difficult to digest.

The authors make a strong claim that they have found substantial differences in binocular interaction between cortical and sub-cortical circuits. But when I look at Figures 1 and 2, which are meant to convey this conclusion, I'm struck by how similar the results are. If the authors want to continue to make their claim, they need to spend more time making the case.

Figure 5 is thankfully easy to understand and shows a very clear result. These perceptual results deviate dramatically from the essentially winner-take-all results for spatial sinewaves shown by Legge & Rubin (1981); whom they should cite by the way. Thus, very interestingly the binocular combination of temporal variation is quite different than the binocular combination of spatial variation. Can the pupil and EEG results also be plotted in the fashion of Figure 5? You'd pick a criterion pupil (or EEG) change and use it to make such plots.

My main suggestion is that the authors need to devote more space to explaining what they've done, what they've found, and how they interpret the data. I suggest therefore that they drop the computational model altogether so that they can concentrate on the experiments. The model could be presented in a future paper.